Analysis plan v1.0: L3 SST reliability and forecast skill, northern Bay of Bengal

Oct 10, 2026 · @Md. Ridwanul Banna Rifat

1 Purpose and what needs approval

This plan fixes every analysis choice before the reliability–skill link is tested, so that the results cannot shape the method. It follows the MS research proposal and states each change from it in Section 5.

Status on 10 October 2026: the forecasting code reproduces Manuscript 1; the 2015–2024 hindcast is done (LSTM with one seed so far); L4 SST and its analysis uncertainty are extracted at the seven stations; iQuam scoping for 2018–2020 is done. No link statistic has been computed. Downloading the L3 data and computing the RQ1 reliability metrics may start before sign-off, because they do not touch forecast errors.

The supervisor is asked to approve:

1. The product set and window (Section 2).
2. The scale for each reliability dimension, which the scoping data decided (Section 3).
3. The primary tests and their decision rules (Section 4).
4. The deviations from the proposal (Section 5).

2 Data, products and units

Item	Choice	Reason
Stations	Seven grid cells of Manuscript 1, Table 1	Same mapping as the forecasts; re-extraction matched published means within 0.004 °C
Station box	20.0–22.5°N, 89.0–92.5°E	Same as Manuscripts 1 and 2
Regional box	18–23°N, 86–93°E	Holds the offshore drifters used for Accuracy

Item	Choice	Reason
Primary L3 product	OSI SAF MetOp-B AVHRR L3C, 0.05°, day and night files (PO.DAAC, AVHRR_SST_METOP_B_GLB-OSISAF-L3C-v1.0)	MetOp AVHRR is an input to the SST CCI v3 record behind the L4 [confirm the MetOp-B period in the CCI Product User Guide]; also in Li et al.'s product list
Second and third L3 products	NOAA ACSPO MODIS Aqua and MODIS Terra L3C, 0.02°, day and night files (NOAA CoastWatch)	The proposal's MODIS pair, in the same ACSPO L3C form Li et al. used; independent of the L4
L3 window	January 2018 to December 2020	NOAA advises against ACSPO MODIS data more than one year after each satellite's last orbit manoeuvre (Terra 27 February 2020, Aqua 18 March 2021); 2018–2020 is inside both limits
In situ reference	iQuam v2.10 drifters, quality level ≥ 4	As in Li et al.; quality level 5 as a check
L4 and uncertainty	C3S-GLO-SST-L4-REP-OBS-SST: analysed_sst and analysis_error_sst	The forecasts use this dataset
Monthly uncertainty u	Mean of daily analysis_error_sst at the station cell	The field holds values near −273 K on 30 March every year (175 station-days); these are excluded
Bathymetry	GEBCO_2023 (ETOPO1 used for scoping only)	Shelf < 200 m, deep ≥ 200 m
Monsoon phases	MAM, JJAS, ON, DJF	Proposal section 6.6
Day and night	Solar zenith angle < 85° day, > 95° night, twilight excluded	Li et al.; proposal section 6.5

The basic unit is product × station × monsoon phase × day/night: 56 units per product, 168 for the three products. Each station unit takes its Accuracy values from the regional unit of the same product, phase and day/night (Section 3).

3 Metrics, scales and screening

What the scoping showed (2018–2020). Nine drifters supplied 5,215 day or night records in 18–23°N, 86–93°E, one of them 36%. None came within 50 km of a station: the closest record was 68 km away, the median 264 km. On the shelf, the post-monsoon and winter held fewer than 50 records per day/night unit; offshore, every unit held 305–807 records. Drifters were present mainly in November–December 2018, April–August 2019 and November 2019 to March 2020, so each phase rests on two to four months. These counts are upper bounds on matchups, since clouds remove more.

Scale for each dimension, decided by the scoping:

Dimension	Metrics (Li et al.)	Scale	Reference	Minimum support
Accuracy	A1 absolute bias, A2 RMSE, A3 r	Regional: offshore (≥ 200 m) part of the regional box × phase × day/night; shelf matchups reported descriptively	Drifters	n ≥ 50 matchups
Availability	B1 mean valid coverage, B2 longest gap, B3 gaps per 90 days	Station: water pixels within 25 km of each station grid cell (10 km as a check)	None	Effective day: coverage ≥ 10%
Error Consistency, station	C1 SD of monthly RMSD	Station, day and night units separately	L4 analysis of the same day (He et al., 2024)	≥ 10 pixel pairs per month, ≥ 2 months per unit
Error Consistency, regional	C2 absolute day–night bias difference	Regional offshore, by phase	Drifters	≥ 50 matchups per side

Why these scales.

- *Availability radius.* At 0.05°, a 5 km radius holds a single pixel, and an L4 value at a cell draws on observations tens of kilometres away. Water pixels are those marked as water in the L4 mask and deeper than 0 m in GEBCO.

- *C2 moves to the regional scale.* The L4 is a daily analysis, so a day–night bias against it at the stations would include diurnal warming as well as retrieval error. Against drifters matched within ±3 h, it does not.
- *C3 is dropped.* It compares shelf and deep water, and all stations are on the shelf. At the stations, the C score is C1 alone; regionally, C2 is reported next to Accuracy.
- *C1 for MetOp-B is not fully independent.* The L4 ingests MetOp AVHRR, so the MetOp-B differences from the L4 are partly circular. The two MODIS products, which the L4 does not use, give the independent check.
- *Accuracy fallback.* If a regional unit has fewer than 50 matchups after cloud screening, its Accuracy window is extended back to 2016–2017 (iQuam scoping repeated for those years); if it is still short, that unit's Accuracy is descriptive only, as proposal section 6.3 allows.

Satellite pixels need GHRSST quality level ⩾ 4 (5 as a check). Matchups use ±3 h and 5 km, with the pixel time taken from sst_dtime; the closest in time is kept, then the closest in distance.

Screening and scoring. A station unit is scored only when its regional Accuracy, its Availability and its C1 are all valid; products are ranked only where at least two are valid; missing metrics are not imputed and weights are not redistributed. Within-dimension weights follow Li et al. (A: 0.25, 0.55, 0.20; B: 0.50, 0.30, 0.20; C at the stations: C1 only). Units that fail are reported as insufficient evidence, not as poor performance.

Within-unit normalised scores rank products inside one unit only, so the link analysis uses raw metrics: MetOp-B AVHRR for the primary test, and the two MODIS products and the three-product mean as secondary.

Checks for RQ1 (added 11 October 2026, after a test of the code on synthetic data, before any real L3 data were processed).

- *Matchup count.* Li et al. keep one matchup per drifter record. Drifters report hourly, so one drifter near one pixel yields up to six records within ±3 h; on synthetic data this inflated n about sixfold. The n ⩾ 50 rule follows Li et al., the count of drifter passes is reported beside it, units with fewer than 10 passes are flagged, and Accuracy is recomputed with one matchup per drifter per satellite file.
- *Score sensitivity.* With two or three products, min–max scaling turns any difference into a 0–100 gap. On synthetic data where only accuracy differed, sampling noise in coverage and C1 changed the winner in more than half of the units. Li et al.'s scaling stays the primary method; a check sets a smallest meaningful difference per metric (A1 and A2: 0.05 K; A3: 0.02; B1: 0.05; B2: 3 days; B3: 1 event per 90 days; C1: 0.05 K), below which products score nearly equal, and reports how many winners change.

4 Forecast errors and tests

Forecast errors. One-month-ahead forecasts of monthly mean L4 SST at the seven stations, rolling origin 2015–2024: for each test year, training runs from January 2000 to December of the previous year (840 station-months per method). Methods: persistence, training climatology, SARIMA(1,1,1)(1,1,1)12 and the Manuscript 1 LSTM, unchanged, averaged over seeds 42–46. Each absolute error is divided by the interannual SD of the same station and calendar month in that fold's training years:

$$z_{s,m,y} = \frac{|e_{s,m,y}|}{\sigma_{s,m}^{\mathrm{train}}}$$

Primary method: SARIMA, which had the lowest error at all seven stations over 2015–2024. The other three methods are secondary.

Primary test 1, direct route (RQ2). Station-month regression of z on the L4 analysis uncertainty u of the target month, with station × calendar-month fixed effects and a linear trend in year:

$$z_{s,m,y} = \alpha_{s,m} + \delta\,(y - 2015) + \beta\, u_{s,m,y} + \varepsilon_{s,m,y}$$

The coefficient β is reported per 0.1 K of uncertainty. Inference: a Freedman–Lane permutation test with 9,999 permutations, in which residuals of the model without u are permuted as whole years within each calendar month for all stations together; this keeps the trend term valid and the dependence between neighbouring stations. The 95% interval comes from a wild cluster bootstrap clustered by year (Webb weights), because an ordinary bootstrap is unreliable with ten clusters.

Interpretation: u of the target month measures how noisy the verifying value is, so β combines a noisier target with weaker forecast inputs (Ferro, 2017). The secondary analysis with u averaged over the 12 input months separates the two in part.

Primary test 2, indirect route (RQ2). Across the 28 station × phase cells: Spearman correlation between MetOp-B AVHRR availability B1 (mean of day and night, 2018–2020) and mean z (SARIMA, 2015–2024), after removing phase means from both. Inference: exact permutation of the seven stations' phase blocks (7! = 5,040), as proposal section 6.10 asks. The association across the four phases themselves is shown as a figure without a test: four points cannot separate data support from the seasonal change in how long SST anomalies persist.

Expected signs: $\beta > 0$ (more uncertainty, larger error); negative correlation in test 2 (more coverage, smaller error). Both tests are two-sided at $\alpha = 0.05$.

Premise check (reported, not a hypothesis test): station-month correlation between B1 and u in 2018–2020 (252 station-months).

Secondary analyses (Benjamini–Hochberg false discovery rate, q = 0.10, within each family):

- the same two tests for LSTM, climatology and persistence;
- the other reliability metrics (B2, B3, C1), Aqua MODIS or Terra MODIS in place of MetOp-B, and the three-product mean;
- u as the RMS of daily values, and u averaged over the 12 input months;
- year fixed effects in place of the linear trend;
- test 2 with 2018–2020 errors only;
- availability radius of 10 km instead of 25 km;
- quality level 5 for satellite pixels and for iQuam.

RQ3: z and the reliability metrics related to monsoon phase, interannual SST SD, seasonal amplitude and the number of effective observation days, one factor at a time and in one joint model chosen before fitting.

RQ3 specification (fixed 13 October 2026, before any Step 5 output existed). The unit is the station × phase cell (28 cells).

- *Factors.* Phase; interannual SD σ (SD across 2000–2024 of monthly L4 SST for each calendar month, averaged over the phase's months, °C); seasonal amplitude A (maximum minus minimum of the 2000–2024 monthly climatology at the station, °C); effective-day share f (share of days in the phase, 2018–2020, with MetOp-B coverage ≥ 10% within 25 km, mean of day and night).
- *One factor at a time (family C, Benjamini–Hochberg q = 0.10).* Spearman correlation after removing phase means, with exact permutation of station blocks (5,040): mean SARIMA z (2015–2024) against σ, A and f; MetOp-B B1 against σ and A (not f, which is built from the same daily coverage); MetOp-B C1 against σ, A and f. Phase means and the share of variance explained by phase are reported for each outcome.
- *Joint model (one, for mean SARIMA z).* OLS across the 28 cells with phase intercepts and the three factors standardised across cells, so each coefficient is in z per SD of its factor. p-values: Freedman–Lane with whole-station blocks, exact (5,040 orderings). 95% intervals: wild cluster bootstrap-t by station, Webb weights, 9,999 draws. Variance inflation factors are reported; a coefficient with VIF > 5 is read as not separable from the others.
- *Confidence table (descriptive).* Per cell: the RQ1 winners for day and night under the operational scenario; satellite support from the most available product's effective-day share (good ≥ 0.50, fair 0.20–0.49, poor < 0.20); forecast confidence from the SARIMA MSSS against climatology over 2015–2024 (above climatology ≥ 0.10, near between −0.10 and 0.10, below ≤ −0.10), with the SARIMA RMSE in °C.

- *Premise check.* B1 for a station-month is the mean of MetOp-B daily coverage, day and night together; the correlation with u is reported raw and after removing station means and station × calendar-month means.

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\bar{z}_{s,p} = \alpha_p + b_\sigma\,\sigma^{*}_{s,p} + b_A\,A^{*}_{s} +
b_f\,f^{*}_{s,p} + \varepsilon_{s,p}
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Reporting rules. Effect sizes and intervals are reported whatever the p-value. A null result is reported as a finding. With seven nearby stations, the effective sample is closer to 120 independent months than to 840, and the paper says so.

Power of primary test 1 (computed 11 October 2026 from the real u series and simulated errors, without any forecast error). Within a station and calendar month, u varies little from year to year (SD 0.025 K). The design detects an effect of 0.3 in z per 0.1 K with about 90% power and 0.2 with about 50%; effects of 0.1 or less are likely to be missed (power below 20%). Without any effect, the test rejected in 5.75% of 400 simulations at α = 0.05, so it keeps its stated error rate. The paper reports β both per 0.1 K and per SD of u, and a null result is read against these power figures.

5 Deviations from the proposal and sign-off

#	Proposal	This plan	Reason
1	MODIS Terra and Aqua daily L3 at 4 km (NASA OBPG), VIIRS if feasible	MetOp-B AVHRR (OSI SAF L3C) as primary; Terra and Aqua MODIS as NOAA ACSPO L3C at 0.02°	AVHRR is an input sensor of the L4, MODIS is not. The 4 km OBPG files are binned and carry no pixel time, so a ±3 h matchup is impossible; ACSPO L3C files carry sst_dtime and quality_level
2	L3 window 2023–2024	2018–2020	NOAA advises against ACSPO MODIS data more than a year after the last orbit manoeuvre (Terra 27 Feb 2020, Aqua 18 Mar 2021)
3	Accuracy at each station	Accuracy for the offshore northern bay, by phase and day/night, used as regional context for all stations	No drifter within 50 km of any station in 2018–2020

#	Proposal	This plan	Reason
4	Error Consistency (C1–C3) against drifters, described in section 12 as satellite-only	C1 against the L4 at the stations; C2 against drifters offshore; C3 dropped	No drifters near the coast; a day–night bias against a daily L4 would include diurnal warming; C3 needs deep water. C1 and C2 always need a reference, so they are not satellite-only
5	Availability area not specified	Water pixels within 25 km of each station cell	A 5 km radius holds one 0.05° pixel
6	LSTM test errors 2023–2024	Rolling origin 2015–2024, four methods	4–8 months per station-phase are too few; one method cannot separate data effects from model effects
7	Normalised scores linked to RMSE	Raw metrics linked to standardised error z	Normalised scores are not comparable across units; raw error follows seasonal predictability
8	L4 uncertainty averaged by station-phase as a supplementary check	L4 uncertainty at station-month level as primary test 1	Same dataset as the forecasts; 840 station-months instead of 28 cells

Changes 1 and 2 use the product-scoping step of proposal section 6.2, and changes 3 and 4 follow the contingencies of sections 6.3 and 12 together with the scoping data in Section 3. Changes 5–8 are design changes that the proposal does not foresee, and they need the supervisor's explicit approval.

Amendments. Any change after sign-off is added below with its date and reason, before the affected analysis is run.

Sign-off

Role	Name	Signature	Date
Supervisor	Prof. Mohammad Muslem Uddin	Approved without changes (reported by the student)	11 Oct 2026
Student	Md. Ridwanul Banna Rifat		

Run log. 11 October 2026: the supervisor approved v1.0 without changes. Primary test 1 and secondary family A were run the same day, after this entry, because they need no Level-3 data. Test 2, family B and the premise check run after Step 5 with the same code and seed. No analysis choice was changed.

Amendment 1 (proposed 11 October 2026 after an external review; approved by the supervisor on 13 October 2026, as reported by the student; made before Step 5 and Test 2 are run). No primary test, threshold or model changes; under item 7 the primary window of Test 2 changes to 2018–2020.

#	Item	Change	Reason
1	Definition of u	u for station s, month m and year y is the mean of daily analysis_error_sst at the station cell over the valid days of that month (28–31 days). The code already computes this; the text now states it.	The earlier wording read as a 2000–2024 average
2	Forecast protocol	Model parameters are estimated once per fold (data to December of the previous year); each month t of the test year is forecast one step ahead from observed SST up to t − 1; no forecast uses an earlier forecast. Code unchanged.	The earlier wording was ambiguous
3	Scales and terms	Accuracy is regional and offshore, not station accuracy. C1 is called analysis-relative consistency. Station scores differ only through Availability and C1.	Avoids implying independent validation at each station
4	Accuracy uncertainty	Add 95% intervals from a bootstrap over drifter passes, leave-one-drifter-out ranges and the share of the largest drifter (descriptive).	The 5,215 in situ records came from 9 drifters, one supplying 36%

#	Item	Change	Reason
5	Resolution check	Add availability with all products on a common 0.05° grid (a cell counts when at least half of its native pixels are valid) and report how rankings change.	AVHRR is 0.05°, MODIS 0.02°
6	Reporting	Raw metrics are the main RQ1 result; scores and rankings are secondary, with their sensitivity. "Prespecified" replaces "pre-registered", and plan v1.0 is archived with a public timestamp (OSF or Zenodo) before Test 2 is run.	Supervisor approval is not a public registration
7	Test 2 window	**Decided: option** B (13 October 2026, after the second external review; set to option A earlier the same day). Mean z over 2018–2020, the L3 window, is primary; mean z over 2015–2024 is secondary (family B). The change was made after Step 5 had produced B1 and before any Test 2 statistic or any z–B1 association was computed. Evidence computed without B1: the station × phase pattern of υ in 2018–2020 matches 2015–2024 (Spearman 0.97), while z from 2018–2020 alone (6–12 months per cell) matches the ten-year pattern only partly (Spearman 0.67). Both versions are reported either way.	The reviewer asked for aligned periods

Test 1 reading. The result is inconclusive: permutation p = 0.042, but the year-clustered interval includes zero and the sign is opposite to the expected one. Exploratory checks: leaving out one year gives β from −0.46 to −0.19 per 0.1 K, leaving out one station from −0.37 to −0.30, and whole-year permutation gives p = 0.037. Note added 13 October 2026: the plan sets no rule for a disagreement between the permutation test and the interval. By the planned test alone, β differs from zero at α = 0.05 (Monte Carlo standard error of p

about 0.002), in the direction opposite to the expected one. Calling the result inconclusive is a judgement made after the result was seen; it rests on the exploratory checks logged below.

Supervisor decision on Amendment 1 and item 7: all seven items approved as proposed (reported by the student). Item 7 was first set to option A because no option was named; after the second external review the student reported that 2018–2020 must be primary, so item 7 is option B. **Date:** 13 October 2026

Run log, 13 October 2026. Amendment 1 approved; item 7 set to option A. Step 4 finished on Colab with no failed files. The RQ3 specification in Section 4 was fixed today, before any Step 5 output existed. From code version 2026-10-13a, Step 6 refuses to run Test 2 until this plan is archived with a public timestamp (item 6) and its DOI or URL is given.

Run log, 13 October 2026, Step 5. Step 5 was run on Colab with code version 2026-10-12a (the Step 5 code is identical in 2026-10-13a): the main run (quality level ⩾ 4, 25 km) and the two checks (quality level 5; 10 km radius). Matchups came from the same nine drifters as the scoping. No forecast error was read by Step 5.

Deviation 1 (execution, 13 October 2026). The GEBCO_2023 server could not be reached from Colab, and Step 5 used its built-in fallback, ETOPO1 (1 arc-minute) from NOAA ERDDAP. The drifter scoping in Section 3 also used ETOPO1, so the shelf and offshore classes agree with it. The change affects only the shelf/offshore split of matchups and the water mask of the station areas; no test, threshold or model changes. Test 2, family B, family C, the premise check and RQ3 have not been run.

Accuracy fallback (Section 3), 13 October 2026. Five offshore units have fewer than 50 matchups in 2018–2020: monsoon day for all three products (MetOp-B 11, Aqua 17, Terra 28) and monsoon night for Aqua (33) and Terra (44). As Section 3 prescribes, their window is extended to the June–September months of 2016–2017 (iQuam scoping repeated for those months; code version 2026-10-13b, step5b_accuracy_fallback.py). All other units keep their 2018–2020 values; a unit still short after the extension is reported as descriptive only.

Run log, 13 October 2026, after the second external review. Item 7 of Amendment 1 is changed to option B: Test 2 uses mean SARIMA z over 2018–2020 as primary and over 2015–2024 as a family B variant. No Test 2 statistic and no association between z and any L3 metric had been computed. RQ3 keeps z over 2015–2024, and its joint model is also reported with z over 2018–2020 beside it. Code version 2026-10-13c; the stage 1 numbers are reproduced exactly.

Exploratory checks of Test 1 (not in the plan; step6d and step6e, 13 October 2026). The two prespecified inferences differ because they treat years differently. The permutation test assumes that, within a calendar month, residuals are exchangeable across years; year-level residual SD ranged from 0.36 (2015) to 0.73 (2022), and 2022 alone carried 36%

of the squared year scores. Inferences that allow years to differ do not reject β = 0: permutation with the cluster-robust t, p = 0.18; restricted wild cluster bootstrap, p = 0.21 (Webb) and 0.22 (Rademacher, all 1,024 sign patterns); cluster-robust t with 9 degrees of freedom, p = 0.16. The inverted permutation interval is −0.66 to −0.02 per 0.1 K with the classical t and −0.88 to 0.18 with the cluster-robust t. Seven other definitions of u (RMS of daily values; monthly-mean uncertainty for independent daily errors and for autocorrelated errors with time scales of 3, 7 and 15 days; log; rank within station and month) give β of −0.08 to −0.09 z per SD of u, permutation p 0.04–0.08 and restricted bootstrap p 0.18–0.25. Test 1 stays inconclusive.

Accuracy fallback result, 13 October 2026. The iQuam scoping and Step 5 were repeated for June–September 2016 and 2017 on Colab (code version 2026-10-13b). The regional box held 1,506 records of quality level 4 or 5 in those months, from one moored buoy, 12 Argo floats and six ships, and no drifter, so no matchup was added. As Section 3 prescribes, the five short units keep their 2018–2020 values and are reported as descriptive only. Metrics, screening, scores and rankings are unchanged.

Correction, 13 October 2026. Recomputing the accuracy intervals from the saved matchups showed that the Colab run of Step 5 read one drifter's id once as a number and once as text, so each winter offshore unit listed five drifters instead of four. Step 5b (code version 2026-10-13d, which also fixes the id reading in Step 5) recomputes the drifter counts and the leave-one-drifter-out ranges; the ranges of three winter units changed by up to 0.06 K. Bootstrap intervals, metrics, scores and rankings did not change. Final RQ1 tables: results/step5_final.